

In the name of God
University of Tehran
Faculty of Electrical and Computer Engineering
Course: Neural Networks and Deep Learning
Homework 6

The following table:

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Rules

Before answering the questions, please read the following items carefully. ²¹

- Prepare a report of your answers in the format provided on the course page in the Elearn system, named REPORTS_TEMPLATE.docx. ²²
- It is recommended to do the assignments in groups of two. More than two people are not allowed, and individual submissions do not receive extra credit. Note that group members do not have to remain the same until the end of the term (i.e., you can do the first assignment with person A and the second with person B, etc.). ²³
- The quality of your report is of special importance in the grading process; therefore, please mention all the points and assumptions you have made in your implementations and calculations in the report. ²⁴
- In your report, use captions for figures and titles for tables, as shown in the provided template. ²⁵
- It is not necessary to provide detailed explanations of the code in the report, but you must report and analyze the results obtained from it. ²⁶
- Analysis of the results is mandatory, even if not explicitly requested in the question. ²⁷

- Teaching assistants are not required to run your codes; therefore, any result or analysis requested in the question must be clearly and completely included in the report. Failure to do so will result in a deduction from the assignment's score.²⁸
- The code must be prepared in a notebook format with the .ipynb extension. At the end, the entire code should be executed, and the output of each cell must be saved in your submitted file. For example, if a cell's output is a plot included in your report, this plot must exist in both the report and the code notebook.²⁹
- In case of cheating, a score of -100 will be recorded for all individuals involved.³⁰
- The only authorized programming language is Python.³¹
- Using pre-written code for the assignments is strictly forbidden. If two groups use a common source and submit similar codes, it will be considered cheating.³²
- The late submission policy is as follows: After the submission deadline, you have a maximum of one week to submit with a penalty. After this one week, your score for that assignment will be zero.³³

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- First three days: no penalty³⁴
- Fourth day: 5% deduction³⁵
- Fifth day: 10% deduction³⁶
- Sixth day: 15% deduction³⁷
- Seventh day: 20% deduction³⁸
- The maximum score that can be obtained for each question is 100. If the total points for a question exceed 100, a score higher than 100 will not be applied.³⁹
 - For example, if the score obtained for question 1 is 105 and the score for question 2 is 95, the final assignment score will be 97.5, not 100.⁴⁰

- Please place the report, codes, and other attachments in a folder with the following name, compress it, and then upload it to the Elearn system. ⁴¹

HW[Number]_[Lastname]_[StudentNumber]_[Lastname]_[StudentNumber].zip ⁴²

Example:

HW1_Ahmadi_810199101_Bagheri_810199102.zip ⁴³

- For groups of two, submission of the assignment by one member is sufficient, but it is recommended that both members upload it. ⁴⁴

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Question 1. Unsupervised Learning and Domain Adaptation using GAN ⁴⁵

1-1 Introduction ⁴⁶

One of the main challenges in the practical application of deep learning is a problem known as Domain Shift. ⁴⁷This situation occurs when a model trained on a source domain dataset experiences a sharp decline in performance when faced with data from another domain with visual differences (lighting conditions, visual style, etc.). ⁴⁸In many real-world applications, labeling new data (e.g., real images) is difficult, expensive, or even impossible, but synthetic or simpler data, such as computer-generated or simulated data, is readily available. ⁴⁹

The paper "Unsupervised Pixel-Level Domain Adaptation with GANs" addresses this exact problem. ⁵⁰The authors of this paper present a solution that uses GANs to transform source domain images into images with the appearance of the target domain, such that the original content of the image is preserved while its appearance resembles the target data. ⁵¹This process is performed without using any labels in the target domain. ⁵²The key advantage of this method is the transfer of knowledge from one domain to another at the pixel level, an approach that not only improves the model's performance in the target domain but also produces interpretable and analyzable output. ⁵³In this exercise, by reconstructing this paper, you will become familiar with advanced concepts like domain adaptation and adversarial learning, and you will also gain practical experience in designing and analyzing the performance of hybrid models. ⁵⁴

1-2 Theoretical Section (30 Points) ⁵⁵

- **1-2-1** Why is GAN training unstable? Name and explain three main factors of instability. What mechanisms do you think have been proposed to counter this instability? (5 points) ⁵⁶
- **1-2-2** In a typical GAN, the Generator transforms only noise z into an image. In this paper, the Generator uses an input image and noise instead. What effect do you think this change has on learning and controlling the output? (5 points) ⁵⁷

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- **1-2-3** The model introduced in this paper includes three main components: a Generator, a Discriminator, and a Classifier. ⁵⁸What role do you think each plays in the learning process? ⁵⁹ How would the overall model's performance change if one of them were removed? (5 points) ⁶⁰
- **1-2-4** One of the special features of the model introduced in this paper is the use of a content similarity loss to preserve content. Explain why preserving the source image content is necessary, and what behavior you would expect from the Generator if this mechanism were removed. (5 points) ⁶¹
- **1-2-5** The model in this paper uses an independent classifier for simultaneous training on the original image and the generated image. What is the advantage of this over training only on generated images? ⁶² Analyze how this choice contributes to learning stability. (5 points) ⁶³
- **1-2-6** The method in this paper is designed for domains like MNIST and MNIST-M, where the difference is in style. ⁶⁴Can this same method be used for domains that have semantic differences, such as different objects, different languages, or different viewpoints? ⁶⁵ Justify your reasoning. (5 points) ⁶⁶

1-3 Practical Section (70 Points) ⁶⁷

Data Preprocessing (5 Points) ⁶⁸

In this exercise, we will use two datasets, MNIST and MNIST-M, which you can download from this link. The MNIST dataset contains black and white images of digits 0 to 9 and is used as the source domain.⁶⁹ The MNIST-M dataset is a modified version of these images, created by adding colored backgrounds, and is considered the target domain. The order of images is preserved in both datasets, meaning that each image in MNIST corresponds exactly to the image with the same index in MNIST-M; the only difference is in the background and color, not the numerical content.⁷⁰

First, select 5 corresponding samples from both datasets using the same index and display the MNIST and MNIST-M images in a row, side-by-side, so that their visual difference and content preservation are observable.⁷¹

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Next, you must prepare the data according to the paper's instructions and place them in separate DataLoaders.⁷² First, convert the MNIST images, which are single-channel (grayscale), to three-channel (RGB) images to match the structure of the MNIST-M data. Both datasets must be resized to the dimensions required by the model architecture (e.g., 32x32). After that, all images must be normalized to the range $[-1, 1]$.⁷³

In the next step, divide each of the two datasets into training and testing sets with a ratio of 80% for training and 20% for testing. Since the order of images is the same in both datasets, the split must be done using common indices to maintain the correspondence between MNIST and MNIST-M samples.⁷⁴ Finally, for each dataset, define two separate DataLoaders, one for training data and one for testing data. In total, four DataLoaders should be created: training and testing for MNIST, and training and testing for MNIST-M. Extract the details regarding preprocessing and batch creation from the paper and implement them accordingly.⁷⁵

Testing the Baseline Model and Observing the Domain Gap (5 Points)⁷⁶

Train a classifier with the exact architecture of the paper's classifier (for MNIST-M) only on the MNIST training data.⁷⁷ Then, evaluate the trained model, without any changes, on the MNIST test data and the entire MNIST-M dataset. Report the accuracy for both. Briefly analyze the difference in accuracy.⁷⁸ Note: The architecture must be identical to the one in the paper.⁷⁹

Model Architecture Implementation (20 Points)⁸⁰

After preparing the data, you must design and implement the three main parts of the model according to the descriptions in the paper.⁸¹ These three parts include a generator for transferring image style, a discriminator for distinguishing between real and fake images, and a classifier for the classification task. The structure of all three models is explained in the paper and must be designed accordingly. Extract the architectural details—number of layers, normalization type, activation function, and data flow in each network—from the text and figures of the paper and implement them in code.⁸²

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Implementation of Cost Functions (10 Points)⁸³

After designing and implementing the model architectures, in this stage, you need to implement the loss functions used in training the model as described in the paper. The paper introduces two main types of loss, each playing a different role in optimizing different parts of the model.⁸⁴

The first is the adversarial loss, which guides the competitive interaction between the Generator and the Discriminator and teaches the Generator how to make its generated images look real from the Discriminator's perspective.⁸⁵ The second is the classification or task loss, which evaluates the Classifier's performance in correctly identifying digits in both the original and the generated images.⁸⁶ You must define these losses separately and use them according to the paper's explanation in the training stages of the Generator, Discriminator, and Classifier.⁸⁷ The coefficients for combining these losses should also be considered as hyperparameters. Extract the exact method of combining these losses during the training flow from the paper and implement it.⁸⁸

Model Training (20 Points)⁸⁹

In this section, you must train the model according to the training structure described in the paper.⁹⁰ The Generator is trained only with source domain images (MNIST); it is given an MNIST image along with noise to produce an image with the appearance of the target domain (MNIST-M) while preserving the numerical content. The Classifier is also trained only on the source domain images, both the original MNIST images and the fake ones generated from them. At no stage of training should the labels from the target domain, MNIST-M, be used. The Discriminator is trained separately and must learn the difference between real MNIST-M images and the generated images.⁹¹

At each epoch, the loss values for the Generator and Discriminator, as well as the accuracy of the Classifier, should be measured on the train and test data and saved, as you will need to display and analyze them in the next stage.⁹² Implement the update order, the method of combining losses, and the data used according to the paper, and train the model. For convenience, you can save the model after each epoch or a certain number of epochs, along with other data, so that you can resume the training process from that point in case of any issues.⁹³

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Important Note on Classifier Implementation⁹⁴

Note that the classifier's behavior is different in the training and testing phases, and this difference must be correctly implemented in the code.⁹⁵ In the training phase, the Classifier is trained only with source domain data (MNIST); this is done solely on real MNIST images and the fake images created from them by the Generator, and only MNIST labels are used.⁹⁶ In this phase, no use of MNIST-M data or its labels is permitted.⁹⁷

In contrast, during the testing phase, the Classifier's performance must be evaluated separately on the MNIST test data (source) and the MNIST-M test data (target).⁹⁸ In this stage, using MNIST-M labels is only permissible for evaluating accuracy, not for learning. This distinction is crucial; any use of target labels during the training phase contradicts the assumption of unsupervised domain adaptation and invalidates the model's results.⁹⁹

Displaying Results (10 Points)¹⁰⁰

In this section, you must display the loss and accuracy results that you saved during training as plots.¹⁰¹

Alongside the plots, you should display a few sample images for a qualitative evaluation of the model. Select five images from the MNIST test set and feed them to the Generator to produce fake images.¹⁰² Then, for each sample, display the following images side-by-side in a corresponding row: the original MNIST image, the image generated by the Generator, and the corresponding real image from MNIST-M. Next to each image, also write the true label and the label predicted by the Classifier. The display order must be maintained to allow for checking whether the Generator correctly changed the image's appearance, preserved the numerical

content, and whether the Classifier correctly identified the digit. ¹⁰³

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Question 2: Reconstruction of Endoscopic Polyp Images with EndoVAE ¹⁰⁴

In this question, you must reconstruct the architecture of the EndoVAE paper and evaluate it using the Kvasir dataset. ¹⁰⁵Note that you will be using the Kvasir dataset instead of the original paper's dataset (KID). ¹⁰⁶

2-1 Data Preprocessing (10 Points) ¹⁰⁷

Download and extract the Kvasir dataset. Then, select the normal images from the following folders:

normal-z-line, normal-pylorus, normal-cecum. ¹⁰⁸Try to use all normal images; using data augmentation to increase the number of normal images can be helpful. ¹⁰⁹

Take the polyp images from the polyps folder in the Kvasir-SEG section. ¹¹⁰

Then, according to the paper, load all images, convert them to the RGB color space, resize each image to 96x96 using bilinear interpolation, and normalize the pixel intensities to the range [0, 1]. ¹¹¹Also, if other preprocessing steps like cropping are appropriate, perform them and provide a justification in the report. ¹¹²Consulting section 4.A of the paper can be helpful for this part. ¹¹³

Next, save the final output in two new folders: ¹¹⁴

- processed/normal: All normal images, dimensions (3, 96, 96) ¹¹⁵
- processed/polyp: All polyp images, dimensions (3, 96, 96) ¹¹⁶

In your report, show at least one example of an image before and after this preprocessing. ¹¹⁷

2-2 Designing the EndoVAE Architecture (20 Points) ¹¹⁸

Perform the implementation according to section three and figure one of the paper.¹¹⁹

- **Encoder:** A convolutional network consisting of six Conv2D+ReLU layers, the output of which is flattened and fed through two FC paths to produce the mu and logvar vectors, each with latent_dim = 6.¹²⁰

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- **Reparameterize function** reparameterize(mu, logvar) as described in the paper.¹²¹
- **Decoder:** First, expand the vector Z to a tensor of size 256×6×6 with an FC layer, and then use seven ConvTranspose2d+ReLU layers. In the final layer, use a Sigmoid function to return the output to a 3×96×96 image.¹²²

Ensure the intermediate dimensions of the layers exactly match figure one of the paper. Also, explain your understanding of each of the important parts of the architecture in your report.

¹²³

2-3 Defining Cost Functions (10 Points)¹²⁴

Define the cost functions according to section three of the paper. Also, write down your understanding of each in the report.¹²⁵

- **Reconstruction Loss:** Use Binary Cross-Entropy (BCE) between X and X'.¹²⁶ You can also test MSE and, in the report, state which one you used with justification or comparison.¹²⁷
- **KL Divergence:** Use the standard VAE formula from the same section of the paper.¹²⁸
- **Total Loss:** The final sum: total_loss = Reconstruction + 1.0 * KL¹²⁹

2-4 Model Training Process (25 Points)¹³⁰

Set the hyperparameters according to section 4.A of the paper.¹³¹ Place the "normal" data in a Dataset and DataLoader with

shuffle=True.¹³²

Perform the training process as described in the paper: ¹³³

1. Model in
train mode. ¹³⁴

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2. Forward pass:
Encoder -> reparameterize -> Decoder. ¹³⁵
3. Calculate Reconstruction Loss and KL Loss and sum them. ¹³⁶
4. backprop and optimizer.step(). ¹³⁷
5. Every few epochs, record the average of each Loss. ¹³⁸

Your report for this section must clearly demonstrate that you have performed the training process correctly and in accordance with the paper. ¹³⁹

2-5 Qualitative Generation and Reconstruction (15 Points) ¹⁴⁰

2-5-1 Generation ¹⁴¹

Load the trained model in
eval mode. ¹⁴² Following section 4.B of the paper, create nine random vectors $Z \sim N(0, I)$. Feed each to the Decoder and obtain the set of 9 generated images. Then, display a 3x3 grid of these 9 images in your report. ¹⁴³

2-5-2 Reconstruction ¹⁴⁴

Select a batch of 5 polyp images from the
processed/polyp folder. ¹⁴⁵ Then, following the paper's description, feed the images to the
Encoder to get

(mu, logvar) and then reconstruct them with the Decoder. ¹⁴⁶

Next, display a 2x5 grid in your report, where each row shows the real polyp image and its reconstructed version. ¹⁴⁷

2-6 Numerical Evaluation (10 Points) ¹⁴⁸

For 50 random images from the processed/polyp folder: ¹⁴⁹

1. Reconstruct each image. ¹⁵⁰
2. Calculate the PSNR and SSIM between the real image and the reconstructed image. ¹⁵¹
3. For those 50 samples, create a table with two columns: "Mean and standard deviation for PSNR" and "SSIM". ¹⁵²

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Note that these metrics are not in the paper, and we are using them for a quantitative analysis of your reconstruction. ¹⁵³

2-7 Analysis of Results and Final Discussion (10 Points) ¹⁵⁴

Qualitative Analysis ¹⁵⁵

- How realistic are the generated images? ¹⁵⁶
- What details does the polyp reconstruction preserve, and what parts are blurred? ¹⁵⁷

Quantitative Analysis ¹⁵⁸

- What do the PSNR and SSIM values indicate about the quality of the reconstruction? ¹⁵⁹
- If the PSNR and SSIM values are low, which part of the architecture, hyperparameters, or preprocessing is likely the problem? ¹⁶⁰

Due to the difference in the dataset, the goal of this implementation is not to achieve results equal to the paper; rather, your understanding of the paper's architecture and your analysis of the results will be the basis for grading. ¹⁶¹

2-8 Bonus Section (5 Points) ¹⁶²

- Define a simple CNN classifier to distinguish between real and reconstructed polyps. ¹⁶³
- Use 200 real polyp images and 200 reconstructed images, divided into train and test sets. ¹⁶⁴
- Calculate its Accuracy and AUC and report them. ¹⁶⁵
- Analyze whether the reconstructed images were similar enough to the real ones for training the classifier. ¹⁶⁶